

LQC Whitepaper

Official Review Edition

Modular Global Liquidity Infrastructure
Connecting Fragmented Web3 Liquidity

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Status legend: Implemented means public repository code/test evidence. Pending means external, deployment-specific or governance evidence is still required. Planned means no completed production module is claimed.

1. Executive Summary

Liquidity Chain (LQC) is a modular infrastructure project for connecting fragmented liquidity across decentralized exchanges and, in later phases, across blockchain ecosystems. The initial engineering focus is an EVM routing stack and LQC Flow automated market maker for controlled BNB Smart Chain testnet validation.

The public repository includes protocol-neutral adapter registration, failure-isolated quotes, gas-aware route comparison, one-to-four-route split optimization, atomic protected execution, native BNB handling, risk limits, emergency controls and testnet deployment validation.

Current status: unaudited testnet MVP. No production deployment, live liquidity, exchange listing, audit approval or guaranteed commercial outcome is claimed.

- Liquidity first - utility second - revenue third - scale last.
- Security gates precede unrestricted deployment.
- Every material claim should be independently verifiable.

2. The Liquidity Fragmentation Problem

Liquidity is distributed across exchanges, pools, fee tiers, networks and assets. A visible quoted price may not represent the best executable result once depth, price impact, pool fees, gas and failure risk are considered.

Users and applications often perform fragmented discovery and submit an order to one venue. This can produce unnecessary slippage, failed execution and inconsistent operational controls.

Fragmentation	Resulting challenge
Many venues	Quotes and interfaces differ
Uneven depth	Price impact varies by order size
Different networks	Assets and messages require separate security assumptions
Operational risk	Paused or compromised routes must be isolated

3. Modular Protocol Architecture

Layer	Responsibility	Status
AMM	Pools, LP accounting, token/BNB swaps	Testnet MVP
Registry	Approved adapters and route state	Implemented
Quote	Failure-isolated comparison	Implemented
Optimization	Capped single/split allocation	Implemented
Execution	Protected atomic swaps	Implemented
Risk	Allowlists, caps, pause controls	Implemented foundation
Lending / bridge	Future protocol modules	Planned

Production configuration, audits, external integrations and live liquidity remain separate gates from repository implementation.

4. LQC Flow AMM

LQC Flow is the native constant-product AMM implementation in the repository. Its factory creates token pairs, each pair holds reserves and issues ERC-20 liquidity shares, and the router coordinates liquidity and swaps.

- Constant-product reserve model with a 0.30% swap fee retained for LPs
- Minimum liquidity permanently locked
- Exact-input and exact-output swaps
- Direct and multi-hop paths
- Token/native-BNB liquidity and swap support
- Deadline and minimum-output protection

LQC Flow is testnet-oriented and unaudited. It must not be used with production funds before independent review and formal launch approval.

5. Adapter and Registry Model

Each reviewed DEX integration uses a protocol-specific adapter. The registry records adapter identity, priority and active state. A failed quote is isolated so another enabled route may still be evaluated.

Protocol-neutral architecture does not mean automatic compatibility. Every integration requires address review, path validation, automated tests and security review before activation.

Integration	Implemented scope
LQC Flow	Native direct/multi-hop quote and execution
PancakeSwap V2	Compatible-router quote and protected exact-input execution
PancakeSwap V3	Packed paths, reviewed fee tiers/pools and maximum three hops

6. Quote and Route Discovery

The quote layer queries enabled adapters independently and excludes failed calls from comparison. Deterministic registry priority resolves ties and makes selection reproducible.

Expected output is only one component of execution quality. The intended comparison considers pool fee, price impact, route gas, minimum output, deadline and route availability.

A quote is an estimate, not a guarantee. Realized output depends on state changes, inclusion timing, network conditions and user-selected protection.

7. Gas-Aware Selection

The gas-cost oracle converts estimated BNB gas into units of the output token. The router can therefore compare estimated net output rather than gross quote alone.

- Primary and secondary feed checks
- Positive-value enforcement
- Freshness windows
- Maximum feed-deviation rule
- Governance-controlled configuration

Production oracle addresses, heartbeat windows, deviation thresholds and fallback policy are not finalized. The current implementation uses test infrastructure and does not guarantee best execution.

8. Split Optimization

The optimizer samples allocations and may distribute one order across two to four eligible routes when the estimated net result is superior. Allocation and expected-output totals must reconcile exactly.

The auto router converts the optimizer result into an atomic execution plan and derives route-level minimum outputs from the user's slippage tolerance.

- One-to-four active route cap
- No duplicate DEX identifiers
- Deterministic allocation and tie behavior
- Inactive legs excluded
- Invalid part and route bounds rejected

9. Protected Atomic Execution

The execution router transfers the specified input, grants an exact temporary approval to the selected adapter and verifies recipient output. Deadline and minimum-output conditions apply to supported exact-input flows.

For split execution, every leg is part of one transaction. If a later adapter fails, earlier transfers, pool changes and risk usage revert.

Repository tests require zero residual router/adapter custody after successful and reverted paths. This tested property is not a substitute for an independent audit.

10. Native BNB Handling

The native router wraps BNB into WBNB before protected token execution and unwraps WBNB for the final native-asset transfer. It is designed not to retain user balances.

- Configured WBNB address
- Controlled receive path
- Minimum-output and deadline enforcement
- Exact approval lifecycle
- No unsolicited native transfer acceptance

Each deployment must verify the canonical WBNB address and module linkage on the selected network.

11. Risk Registry

The risk registry stages exposure through token allowlisting and input caps. Limits apply before adapter execution and rejected transactions must not consume daily capacity.

Control	Purpose
Token allowlist	Restrict eligible input assets
Per-transaction cap	Bound single-order exposure
Per-DEX/token cap	Limit venue-specific exposure
UTC-day cap	Bound aggregate daily input
Module pause	Stop new protected execution

12. Governance and Emergency Roles

Structural expansion is reserved for timelocked governance. Guardians may disable a DEX or pause new swaps immediately, but they cannot resume service, move user funds, add adapters or increase limits.

A designated risk role may reduce existing limits but cannot create a new permission or expand exposure. Emergency-controller ownership uses nomination and explicit acceptance.

Final production multisig addresses, signer threshold, timelock delay and operating procedures remain pending external evidence.

13. Token Compatibility Boundaries

The current execution design supports standard ERC-20 accounting assumptions. Fee-on-transfer inputs are explicitly rejected when the received balance differs from the requested input.

- Fee-on-transfer execution: unsupported and rejected
- Rebasing token execution: unsupported
- Permit signatures: deferred
- Non-standard approval behavior: integration review required
- Token and pool allowlists: production decision pending

Unsupported behavior must not leave retained balances, approvals or risk-accounting residue.

14. Reproducible Engineering Baseline

At the publication baseline, the repository compiles 35 Solidity source files and reports 53 passing automated tests using locked dependencies. The exact Git commit and its CI run are authoritative when the suite evolves.

Area	Representative evidence
AMM	Reserves, product, LP lock, liquidity and swap flows
Router	Quotes, adapters, gas, split and rollback
Risk	Allowlists, caps, model-based daily accounting
Governance	Timelock, pause-only guardian, ownership transfer
Deployment	Chain 97, bytecode, ownership and module links

Passing tests do not prove the absence of vulnerabilities or certify production economic safety.

15. BSC Testnet Deployment

Deployment tooling targets BSC testnet chain ID 97. It deploys test tokens, LQC Flow and Router 2.0 modules, establishes ownership and registers the native adapter. Reviewed PancakeSwap endpoints may be supplied separately.

- Reject unexpected chain IDs
- Record deployed contracts and external dependencies
- Record compiler and optimizer settings
- Record governance and adapter configuration
- Never commit deployer private keys

A generated deployment record is evidence only when its addresses and transactions are independently checked on the explorer.

16. Address and Explorer Verification

The read-only validator checks deployed bytecode, Router-to-Factory and WBNB relationships, registry order, adapter status, module linkage, ownership and minimum timelock delay.

The verification-package generator produces Solidity Standard JSON input, compiler settings, constructor arguments, deployed addresses and the recorded source revision for BscScan submission.

Explorer verification remains pending until a real deployment record is validated and published against the exact reviewed commit.

17. Lending Design Direction

Lending is a future module and is not part of the current production claim. The design direction covers approved collateral deposits, borrowing, repayment, health-factor monitoring and controlled liquidation.

Parameter	Current design disclosure
Maximum LTV	45% baseline
Liquidation threshold	60%
Base liquidation penalty	5%
Automatic additional borrowing	Off by default; explicit opt-in required

Final values require economic simulation, oracle validation, liquidity testing, audit, legal review and formal governance approval.

18. Oracle and Liquidation Principles

Account risk should be determined by collateral value, debt and health factor rather than price movement alone. A shallow LQC pool must not be the sole lending oracle.

- Validated external feeds
- Sufficiently liquid DEX TWAP where appropriate
- Freshness and deviation checks
- Debt and market-specific caps
- Partial liquidation where safe and economical
- Emergency pause and recovery procedures

Production liquidation requires adversarial simulation, keeper design, bad-debt handling and clearly disclosed parameters.

19. Cross-Chain Direction

Cross-chain expansion is phased future scope. Non-EVM venues are not assumed compatible with EVM adapters and must use a separately reviewed messaging and accounting layer.

- Authenticated messages
- Nonce consumption and replay protection
- Rate limits
- Emergency pause
- Canonical asset identification
- Global supply reconciliation

Cross-chain movement must not create unexplained LQC supply. Bridge design and audit are separate from Router 2.0 review.

20. LQC Utility Boundaries

The approved design direction associates LQC with future loan-origination and repayment fee settlement and permanent burning of those fees. This does not mean the lending or burn contracts are currently live.

Routing and AMM development does not by itself activate staking, governance, fee benefits, gas support, treasury distributions or other proposed utilities.

Utility activates only after contract completion, audit, parameters, legal review, formal approval and on-chain disclosure.

21. Approved Tokenomics Overview

The current approved project design uses a fixed maximum supply of 1,000,000,000 LQC and planned TGE circulation of 150,000,000 LQC (15%). Earlier draft figures are superseded.

Allocation	Share	LQC
Future ecosystem rewards	35%	350,000,000
Community initial	20%	200,000,000
Team and core contributors	20%	200,000,000
Protocol treasury	10%	100,000,000
Liquidity and market making	10%	100,000,000
Grants and strategic ecosystem	5%	50,000,000

22. TGE and Release Framework

Allocation	TGE / release framework
Ecosystem rewards	0 at TGE; 7+ years, up to 50M annually
Community initial	80M at TGE; remaining 120M activity-based over 24 months
Team / contributors	0 at TGE; 12-month cliff, then 36-month monthly vesting
Protocol treasury	10M at TGE; remaining 90M under 5-year budget framework
Liquidity / MM	50M at TGE; remaining 50M linked to exchange and pool growth
Grants / strategic	10M at TGE; remaining 40M milestone-based

The planned 150M TGE circulation is Community 80M, Liquidity/MM 50M, Treasury 10M and Grants 10M. Contract and wallet evidence remains pending.

23. Supply and Vesting Verification

Before launch, the token contract cap, total supply, decimals and mint/burn powers must reconcile to the approved disclosure. Every allocation wallet and vesting contract should be labeled and independently verifiable.

- Canonical network and token address
- Deployment and ownership transactions
- Explorer-verified source and ABI
- Mint, pause, blacklist, upgrade and recovery powers
- Beneficiaries, cliffs and vesting schedules
- Circulating-supply methodology and excluded wallets

No on-chain production token evidence is claimed in this edition.

24. Fees, Treasury and Burn

Potential revenue sources include routing and swap fees, future lending and repayment fees, liquidation fees, cross-chain services and Liquidity-as-a-Service arrangements. Actual revenue depends on deployed products and usage.

Fee rates, settlement, conversion, burn addresses, treasury allocation and distributions are not final. A future burn must be verifiable on-chain and permanently remove the relevant LQC from circulation.

No yield, revenue, token-price appreciation or investment return is promised.

25. Development Status

Workstream	Current status
LQC Flow AMM	Implemented testnet MVP
Router 2.0	Implemented testnet MVP
Pancake V2/V3 adapters	Implemented; production review pending
Risk/governance foundation	Implemented; final roles pending
BSC testnet tooling	Implemented; real evidence pending
Lending, bridge, perpetuals	Future separate scope
Independent audit	Not completed

Repository implementation is not equivalent to production launch or commercial adoption.

26. Phased Delivery Plan

Phase	Gate
Foundation	AMM, Router, registry, access and test tooling
Controlled testnet	Addresses, verification, smoke tests and capped routes
Independent review	Audit, remediation, retest and known-issues disclosure
Capped pilot	Final multisig, oracle, monitoring, legal and liquidity approval
Expansion	Measured reliability, depth and incident-free operation
Future modules	Separate design, implementation, testing and audits

Dates should not override security, legal, liquidity or operational readiness gates.

27. Audit and Incident Readiness

An independent audit engagement must pin the exact commit, compiler, settings, contracts, addresses and configuration reviewed. Every Critical and High finding requires documented remediation and auditor retest or an explicit unresolved disposition.

- Named incident responders and escalation contacts
- Alert coverage and pause procedure
- Compromised-key and signer-loss procedure
- Public vulnerability reporting policy
- Postmortem and recovery policy
- Periodic emergency drills

The repository's audit scope and handoff documents are preparation materials, not an audit report.

28. CEX Listing Evidence

Evidence group	Required before submission
Technical	Pinned build, tests, audit, verified source and addresses
Token	Supply, decimals, powers, vesting and circulation proof
Market	Timestamped liquidity, holders, volume and methodology
Operations	Deposits, withdrawals, confirmations, monitoring and contacts
Legal	Entity, KYB/KYC, legal analysis, sanctions and AML controls
Integrity	Market-making policy and wash-trading prohibition

Pending evidence must be completed, marked not applicable with accepted reasoning, or disclosed as unresolved risk. No exchange approval is claimed.

29. Entity, Compliance and Disclaimer

The disclosed project entity is MMXlabs&LQC; LLC, a Wyoming limited liability company. Formation, good standing, managers, beneficial ownership, authorized signatory, KYB and legal-opinion materials must be supplied through the reviewer's secure process.

This document is technical and informational. It is not investment, legal, financial or tax advice; an offer or solicitation; or a guarantee of launch, listing, liquidity, revenue or return.

Digital assets and DeFi involve smart-contract, oracle, bridge, liquidity, market, operational, cybersecurity and regulatory risks.

30. Conclusion and Immediate Priorities

LQC has progressed from architecture into a testable EVM routing stack with a native AMM, external DEX adapters, gas-aware comparison, atomic split execution and staged risk controls.

Priority	Next evidence milestone
1	Validate a controlled BSC testnet deployment and publish matching explorer source
2	Finalize multisig, roles, oracle parameters, allowlists and operational limits
3	Commission an independent audit; remediate and retest findings
4	Complete token, vesting, circulation, legal and CEX evidence packages
5	Run a monitored capped-liquidity pilot before any expansion

The next milestone is verified testnet and audit readiness - not unrestricted production deployment.